MILITARY SPECIFICATION

MULTIPLEXER, PULSE-CODE MODULATION

This specification is approved for use by Naval Air Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE.

1.1 Scope. This specification defines the performance and test requirements for a pulse-code modulation (PCM) multiplexer referred to herein as the multiplexer.

2. APPLICABLE DOCUMENTS.

2.1 Issues of documents. The following documents of the issue in effect (unless otherwise indicated) on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-C-45662	Calibration System Requirements.
MIL-H-46855	Human Engineering Requirements for Military Systems, Equipment and Facilities.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commanding Officer, Naval Air Engineering Center, Engineering Specifications and Standards Department (ESSD) Code 93, Lakehurst, NJ 08733, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 5821

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Naval Air Systems Command
(Code Ident 30003)

WS-6536	Procedures and Requirements for Preparation and Soldering of Electrical Connections.

STANDARDS

MILITARY

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-109	Quality Assurance Terms and Definitions.
MIL-STD-129	Marking for Shipment and Storage.
MIL-STD-130	Identification Marking of US Military Property.
MIL-STD-143B	Standards and Specifications, Order of Precedence for the Selection of.
MIL-STD-454	Standard General Requirements for Electronic Equipment.
MIL-STD-461	Electromagnetic Interference Characteristics Requirements for Equipment.
MIL-STD-462	Electromagnetic Interference Characteristics, Measurement of.
MIL-STD-810B	Environmental Test Methods.
MIL-STD-810C	Environmental Test Methods.
MIL-STD-1472	Human Engineering Design Criteria for Military Systems, Equipment and Facilities.

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MIL-STD-1695 — Environments, Working, Minimum Standards for.

(Copies of specifications, standards, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

Federal Cataloging Handbook

H4 — Federal Supply Code for Manufacturers, United States and Canada.

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, D.C. 20402.)

Inter-range Instrumentation Group
(IRIG) Range Commanders Council
(Code Ident 29669)

Document 106 — Telemetry Standards.

(Application for copies should be addressed to the Secretariat, Range Commanders Council, White Sands Missile Range, New Mexico 88002.)

3. REQUIREMENTS.

3.1 Item definition. The device defined by this specification is a digital data multiplexer which accepts analog and parallel digital inputs, and converts those inputs to a serial digital output of the filtered or unfiltered nonreturn-to-zero level (NRZL) format. The multiplexer uses a programmable read-only memory (PROM) to establish the desired multiplex routine for specific project requirements. The number and mixture of inputs

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varies with intended use, as does the output signaling rate, specified in the contract or purchase order (see 6.2.1). The multiplexer, as specified herein, meets all applicable requirements of IRIG Document 106.

3.2 Characteristics.

3.2.1 Performance requirements. The multiplexer shall meet the following performance requirements when connected to input and power supply voltages as specified herein, and produce both a filtered and an unfiltered nonreturn-to-zero (NRZ) PCM bitstream. The multiplexer shall meet all applicable requirements of IRIG Document 106.

3.2.1.1 Input characteristics. The multiplexer shall accept both analog and digital inputs, in the numbers specified in the contract or purchase order (see 6.2.1).

3.2.1.1.1 High level analog inputs. Unless otherwise specified in the contract or purchase order (see 6.2.1), high-level analog inputs shall accept input voltages over the range from 0 to +5 volts direct current (Vdc).

3.2.1.1.1.1 Zero scale. Unless otherwise specified in the contract or purchase order (see 6.2.1), an input voltage of zero volt (V) shall produce an output within 2 bits of 0000001100 when the multiplexer is in a 10-bit output mode.

3.2.1.1.1.2 Full scale. Unless otherwise specified in the contract or purchase order (see 6.2.1), an input voltage of 5.000 Vdc shall produce an output within 2 bits of 1111110100 when the multiplexer is in a 10-bit output mode.

3.2.1.1.1.3 Input resistance. Input resistance shall be sufficiently high that a change in source impedance from 0 to 1 kilohm will not cause a conversion difference in excess of 2 parts in 1,024.

3.2.1.1.1.4 Back current. Back current shall not exceed 10 nanoamperes (nA) at -40 degrees Celsius (°C) (-40 degrees Fahrenheit (°F)) nor 500 nA up to +80°C (+176°F).

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3.2.1.1.1.5 Out-of-band voltages. The multiplexer shall not be damaged nor other channels affected by an out-of-band voltage applied to any input or combination of at least three inputs over the range of ±30 Vdc.

3.2.1.1.1.6 Ground loop current, data ground. Inputs connected to data ground shall read as zero voltage inputs (00000011XX in a 10-bit mode), where X is a "0" or a "1", regardless of voltages on other inputs (within limits of 3.2.1.1.1.5) or the difference in voltage between data ground and power ground (within the limits of 3.2.1.3.3).

3.2.1.1.2 Low-level analog inputs. Low-level analog inputs shall accept differential voltages over the ranges specified in the contract or purchase order (see 6.2.1).

3.2.1.1.2.1 Zero scale. The output code of the low-level analog inputs for minimum (most negative) input voltage shall be within 4 least significant bits (LSBs) of 0000001100 in a 10-bit mode.

3.2.1.1.2.2 Full scale. Output of the low-level analog inputs for most positive specified input voltage shall be within 4 LSBs of 1111110100 in a 10-bit mode.

3.2.1.1.2.3 Common-mode rejection. Common-mode rejection with respect to data ground shall be a minimum of 60 decibels (dB) from direct current to 60 hertz (Hz), and at least 40 dB from above 60 Hz to 1 kilohertz.

3.2.1.1.2.4 Common-mode voltage range. Input common-mode voltage range shall be ±10 Vdc with no degradation beyond the levels specified in 3.2.1.1.2.3.

3.2.1.1.2.5 Input resistance. Input resistance shall be sufficiently high that a change in source impedance from 0 to 1 kilohm will not cause a conversion difference in excess of 4 bits in 1,024.

3.2.1.1.2.6 Back current. Back current shall not exceed 100 nA from either leg to signal ground nor 50 nA differentially in either polarity.

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3.2.1.1.2.7 Out-of-band voltages. Differential inputs shall not be damaged by input voltages of +10 Vdc and -5 Vdc applied differentially.

3.2.1.1.2.8 Recovery time. Inputs shall recover from an out-of-band condition within 1 millisecond (ms) of the removal of such an input. This condition shall apply even when the out-of-band input is present at an unselected low-level analog input.

3.2.1.1.3 Discrete inputs. Unless otherwise specified in the contract or purchase order (see 6.2.1), digital inputs shall produce a "1" at the output for a voltage greater than +2.2 Vdc at the input, and a "0" for a voltage less than +1.2 Vdc at the input. Specified switching threshold if different from the above shall be within the limits of +1 to +9 V, with a band of uncertainty not less than 1 V.

3.2.1.1.3.1 Input impedance. Input resistance shall be no less than 10 kilohms referred to threshold voltage, shunted by no more than 100 picofarads (pF).

3.2.1.1.3.2 Back current. Back current referred to ground or threshold voltage shall not exceed 1 microampere (μA) in either polarity.

3.2.1.1.3.3 Out-of-band voltages. The digital input channels shall operate normally with input voltages between the ranges of ±30 Vdc.

3.2.1.1.3.4 Throughput delay. Response of the output to a change of state at the input shall be correct for input changes as recent as 2 microseconds (μs) prior to selection of that channel.

3.2.1.2 Data transfer characteristics. The multiplexer shall contain the appropriate amplifiers and analog-to-digital converters to produce the desired digital output signals.

3.2.1.2.1 Conversion accuracy. Conversion accuracy for a high-level analog input shall be within 0.2 percent ±0.5 LSB for any combination of noise, crosstalk, channel scatter, and drift for all inputs of 1 kilohm source impedance or less at all environmental temperatures. Conversion accuracy for all low-level analog inputs shall be within 0.4 percent ±0.5 LSB under the same conditions.

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3.2.1.2.2 Resolution. Unless otherwise specified in the contract or purchase order (see 6.2.1), resolution of the analog-to-digital conversion shall be 10 bits (1 part in 1,024).

3.2.1.2.3 Conversion speed. Unless otherwise specified in the contract or purchase order (see 6.2.1), conversion speed shall be 666 nanoseconds (ns) or less for the most significant digit and 10 µs or less for the least significant digit (1 part in 2^{10}).

3.2.1.2.4 Linearity. Linearity shall be within 1 bit in a 10-bit conversion.

3.2.1.2.5 Monotonicity. All conversions shall be monotonic.

3.2.1.3 Power.

3.2.1.3.1 Operating voltage range. The multiplexer shall operate on 28 ±6 Vdc.

3.2.1.3.2 Isolation. Power supply positive and return lines shall be isolated from data ground by a minimum of 1 megohm.

3.2.1.3.3 Differential voltage. The multiplexer shall operate within tolerances with voltage differentials up to and including ±40 Vdc between data ground and either power supply lead.

3.2.1.3.4 Current drain. Unless otherwise specified in the contract or purchase order (see 6.2.1), the multiplexer shall not draw more than 350 milliamperes (mA) from primary power.

3.2.1.3.5 Overvoltage. The multiplexer shall not be damaged by a power supply voltage of +40 Vdc for a period of 1 minute.

3.2.1.3.6 Power reversal. The multiplexer shall not be damaged by supply voltage of improper polarity for an infinite period.

3.2.1.4 Timing circuits.

3.2.1.4.1 Internal clock. The multiplexer shall be provided with an internal clock of frequency specified in the contract or purchase order (see 6.2.1). The clock output shall be available exclusively at the input connector and shall operate the multiplexer only upon connection to the clock input circuit. This clock signal may be at a multiple of the output signal.

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3.2.1.4.2 Clock format. The clock signal shall be a transistor-transistor-logic (TTL) or a complimentary metal-oxide-semiconductor (CMOS) signal driven from an internal +5-Vdc supply referenced to data ground.

3.2.1.4.3 Clock load. The clock shall continue to operate when loaded by an external load of 1.5 kilohms connected to ground or the +5-Vdc supply.

3.2.1.4.4 External clock . The multiplexer shall be capable of being driven by an external TTL or appropriately buffered +5-Vdc CMOS signal at rates from zero to the limit specified in 3.2.1.4.1.

3.2.1.4.5 External clock loading. The load seen by an external clock signal shall not exceed 3 kilohms to ground or the +5-Vdc supply.

3.2.1.4.6 Clock buffer. When specified in the contract or purchase order (see 6.2.1), the multiplexer shall provide a clock buffer circuit which isolates an incoming clock signal, buffers the clock signal through a minimum of four Schottky-clamped TTL inverters, and feeds the resulting buffered clock signal into the normal clock input terminal of the multiplexer through an internal connection. Additionally, the clock buffer shall provide a similarly buffered signal at a separate coaxial connector for driving external equipment. Each string of inverters shall be powered by a separate power supply regulated separately from the multiplexer's other internal supplies and from each other. Unless otherwise specified, the output of the delayed signal at the coaxial connector shall rise between 30 and 80 ns following onset of output data.

3.2.1.5 Output characteristics. The multiplexer shall provide a TTL-compatible NRZL output consisting of a serial digital representation of the analog inputs, most-significant bit first; the digital inputs; a synchronization word; and, when specified in the contract or purchase order (see 6.2.1), a single parity bit at the end of each data word so the number of "1s" in each word will always be odd in accordance with the program directed by the PROM described in 3.2.1.7.

3.2.1.5.1 Output load. The TTL output shall operate normally into a 3-kilohm load connected to data ground or to +5 Vdc.

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3.2.1.5.2 Output short-circuit protection. The TTL output shall not be damaged by a continuous short circuit.

3.2.1.6 Filtered output. When specified in the contract or purchase order (see 6.2.1) a second output shall be provided, consisting of a 2-volt peak-to-peak (V_{p-p}) (4 V_{p-p} open circuit) signal identical to the TTL output described in 3.2.1.5 but passed through a filter at the frequency specified in 3.2.1.4.1 with a constant delay (linear phase) characteristic and a minimum rolloff of 36 dB per octave in accordance with IRIG Document 106.

3.2.1.6.1 Output impedance. Output impedance of the filtered output shall be 75 ohms at the defined clock rate.

3.2.1.7 Programmability. The multiplexer shall be field-programmable by interchange of a PROM. The type and number of devices used for the PROM is not fixed, but the PROM shall control the content and length of the synchronization word, subframe routines, the order in which the inputs are read, the number of bits per analog input (all words shall be of the same length), and the frame length. The PROM may, at the contractor's option, control other features not specified herein. When specified in the contract or purchase order,instructions for programming of the PROM shall be supplied with the multiplexer (see 6.2.2). Field replacement of the PROM shall be accomplished without soldering.

3.2.1.7.1 Subcommutation capability. The multiplexer shall be programmable by the PROM to permit subcommutation of 2, 4, 8, or 16 analog signals or digital words in a single major frame. Subframe identification shall be programmable.

3.2.1.7.2 Synchronization word length. The synchronization word shall be programmable both in content and length from a minimum of one to a maximum of three words.

3.2.1.8 Self-start. The multiplexer shall commence operation within the limits described herein within 100 ms of the application of power and without external triggering at any combination of power supply voltage, input conditions, and environmental conditions within the limits required by this specification. Additionally, the multiplexer shall commence operation following a 2-hour soak, with power removed, at temperatures down to the minimum storage temperature (see 3.2.5.2). Operation need not be within prescribed limits between minimum storage and operation temperature.

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3.2.1.9 Input protection, power off. No input shall be damaged by an input voltage within the limits required by this specification when input voltages are applied to the multiplexer without primary power applied to the multiplexer.

3.2.2 Physical characteristics.

3.2.2.1 Conformance to documents. The multiplexer shall conform to figure 1 or as otherwise required by the contract or purchase order (see 6.2.1), and shall meet all of the requirements of this specification and the documents referenced to the extent specified herein.

3.2.2.1.1 External adjustments. No external adjustments shall be available on the multiplexer; however, the PROM shall be accessible and removable from under a cover plate without otherwise dismantling the unit (see figure 1).

3.2.3 Reliability.

3.2.3.1 Shelf life. The multiplexer shall be designed to meet the performance requirements specified herein after a storage period of up to 5 years.

3.2.3.2 Life. The multiplexer shall have a total operating life of 2,000 hours minimum without necessity for repair, replacement, or adjustment.

3.2.4 Maintainability. Except for the interchange of PROMs (see 3.2.1.7), the multiplexer is not characterized as field repairable. It is, however, a desirable feature that the multiplexer be composed of replaceable modules containing groups of input circuits, the power supply, the analog-to-digital converter, and the output stages.

3.2.5 Environmental conditions. The multiplexer shall meet all requirements specified herein before, during, and after exposure to the following environmental conditions or any natural combinations thereof:

(a) Temperature-altitude: Reduced barometric pressure equivalent to 110,000 feet altitude at a temperature of -40°C (-40°F).

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(b) Storage temperature: Temperature ranging from +85 to -54°C (+185 to -65.2°F).

(c) Operating temperatures: Temperatures ranging from +80 to -40°C (+176 to -40°F).

(d) Temperature shock: Transitions within 5 minutes from temperature stabilization at +80 and -40°C (+176 and -40°F) to an environment stabilized at the opposite temperature extreme.

(e) Humidity: Atmosphere containing 85 percent or greater relative humidity at temperatures from +28 to +71°C (+82.4 to +159.8°F).

(f) Vibration: Levels of sinusoidal and random vibration up to and including 20 gravity units (g) peak and 0.3 g^2, respectively. Temperature levels shall be as specified in 3.2.5(c). No vibration isolators shall be used.

(g) Shock: Levels of 100-g acceleration sawtooth pulses with a duration of 6 ms, in all axes and in any direction. No shock isolators shall be used.

(h) Burn-in and screening: Each multiplexer shall be subjected to a minimum of 6 cycles of burn-in and screening as shown in figure 2.

3.3 Design and construction.

3.3.1 Materials. All materials used in the construction of the multiplexer shall be in accordance with MIL-STD-454, Requirement 3, Flammable Materials; Requirement 4, Fungus-Inert Materials; Requirement 15, Ferrous Alloys, Corrosion Resistance; and Requirement 16, Dissimilar Metals.

3.3.2 Electromagnetic radiation. Unless waived in writing by the contracting officer, the multiplexer shall not generate electromagnetic signals either through radiation, power-line

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conduction, amplitude modulation or phase modulation of the output signal, nor through spikes fed back to inputs selected or not selected. Maximum permissible emission levels shall be as specified in the contract or purchase order (see 6.2.1). Unless otherwise specified in the contract or purchase order (see 6.2.1), the contractor shall perform or have performed testing to MIL-STD-461 and MIL-STD-462, Tests CE01, CE02, CE03, CE04, CE05, RE01, RE02, and RE05. When specified in the contract or purchase order, the results of these tests shall be submitted with each preproduction lot (see 6.2.2).

3.3.3 Identification and marking. The multiplexer shall be identified and marked in accordance with MIL-STD-130 and this specification. Marking shall include, but shall not be limited to the following:

(a) This specification number.

(b) Contract or purchase order number.

(c) Manufacturer's name and code identification from Federal Cataloging Handbook H4.

(d) A unique serial number.

3.3.4 Workmanship. The multiplexer, including all parts and accessories, shall be constructed and finished in a manner that will ensure compliance with all requirements of this document. Except where there is conflict with the applicable drawings or unless otherwise specified herein, fabrication and assembly practices shall be in accordance with MIL-STD-454, Requirement 9. Processes relative to soldering shall be in accordance with WS-6536.

3.3.5 Interchangeability. The multiplexer interchangeability shall be in accordance with MIL-STD-454, Requirement 7.

3.3.6 Safety. A multiplexer produced in accordance with this specification shall present no safety hazard to operating personnel.

3.3.7 Human performance/human engineering. The multiplexer shall meet the requirements of MIL-STD-1472 and MIL-H-46855.

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3.3.8 Standards of manufacture.

3.3.8.1 Production processing. The multiplexer shall be manufactured, examined, and tested under conditions and procedures meeting the following requirements.

3.3.8.1.1 Special working environment. The contractor shall provide adequate facilities for the fabrication, assembly, and testing of the multiplexer to be delivered in accordance with this specification. Unless otherwise specified, the minimum standards for working environments of MIL-STD-1695(AS) shall apply.

3.3.8.1.2 Inspection and test planning. The contractor's program shall provide the necessary planning function for tests and inspections conducted during the entire phase of fabrication, processing, and assembly. The planning shall be based on a comprehensive study of the articles; the fabrication and processing operations; the methods of material integration, assembly, and checkout; and the final test and inspection procedures. Inspections shall be established at points which will minimize delays resulting from deficiencies, and in all cases shall be at or before the last point at which the acceptability of the operation or quality of the characteristics may be verified.

3.3.8.1.3 Process control procedures. Process control procedures shall be prepared when necessary to supplement applicable process specifications to provide detailed performance and control methods. These procedures shall document the preparation; fabrication details; conditions to be maintained during each phase of the process; the methods of verifying the adequacy of processing materials, solutions, equipment, and their associated control parameters, including statistical quality control plans where applicable; and the required records to indicate the results of such inspection and process verification. The contractor's quality organization shall review the written procedures for those process controls and conduct audits to determine that the actual operations conform with approved methods and procedures.

3.3.8.1.4 Material control. Material controls shall ensure that only conforming materials and articles are used. Materials and articles not conforming to or not required for the operation involved shall be removed from the work operations. Positive action shall be taken to protect controlled processes or

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operations from contamination by residue from nonconforming materials and from previous operations. The contractor shall ensure that each operation of inspection (and to the extent practicable, fabrication) is traceable to the individual responsible for its accomplishment.

3.4 First article.

3.4.1 Preproduction sample. Unless otherwise specified in the contract or purchase order (see 6.2.1), a preproduction sample consisting of four multiplexers shall be delivered to the testing activity. The preproduction sample shall be manufactured using the same methods, materials, processes, and procedures proposed for production. Any production prior to acceptance of the preproduction sample shall be at the risk of the contractor.

3.5 Precedence. Specifications, standards, and drawings shall be used in the order of precedence established in MIL-STD-143B. When the requirement of this specification and applicable subsidiary documents are in conflict, the following precedence shall apply:

(a) Contract or purchase order. The contract or purchase order shall have precedence over any other document.

(b) This specification. This specification shall have precedence over all applicable subsidiary documents.

(c) Reference documents. Any document referenced in this specification shall have precedence over all applicable subsidiary documents referenced therein.

4. QUALITY ASSURANCE PROVISIONS.

4.1 General.

4.1.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order (see 6.2.1), the contractor is responsible for the performance of all inspection requirements as

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specified herein. Except as otherwise specified in the contract or purchase order (see 6.2.1), the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this document where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirement.

4.1.1.1 Classification of inspection. The inspection and testing requirements for the multiplexer shall be as follows:

(a) Preproduction inspection (see 4.1.2.1).

(b) Quality conformance inspection (see 4.2).

(1) Individual inspections (see 4.2.2).

(2) Sampling tests (see 4.2.3).

4.1.2 Special tests and examinations.

4.1.2.1 Preproduction inspection. Preproduction inspection shall consist of the examinations and tests of table I and shall be performed in the sequence specified therein. The sample shall be in accordance with 3.4.1.

4.1.2.1.1 Preproduction sample designation. The four preproduction sample multiplexers shall be numbered 1, 2, 3, and 4. Each numbered multiplexer shall be subjected to the examinations and tests delineated in table I.

4.1.2.1.2 Acceptance criteria. Failure of any multiplexer to pass any of the preproduction examinations and tests specified herein shall reject the preproduction sample.

4.1.2.1.3 Disposition of preproduction sample. Preproduction samples shall not subsequently be offered as part of any production lot.

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4.1.2.1.4 Post tests. Post tests shall consist of verification of the following:

(a) Input current (see 4.5.2).

(b) Output waveform (see 4.5.31).

(c) Clock frequency (see 4.5.27).

(d) Zero-scale, high-level inputs (see 4.5.1).

(e) Full-scale, high-level inputs (see 4.5.1).

(f) Zero-scale, low-level inputs (see 4.5.6).

(g) Full-scale, low-level inputs (see 4.5.7).

(h) Threshold voltage, digital inputs (see 4.5.16).

TABLE I. Preproduction inspection.

Examination or test	Requirement paragraph	Method paragraph
Group I (all multiplexers)		
Visual and mechanical	3.2.2.1, 3.3.3, 3.3.4	4.4
Burn-in and screening	3.2.5(h)	4.6.8
Performance tests	3.2.1.1 to 3.2.1.6	4.5.1 to 4.5.34
Temperature-altitude	3.2.5(a)	4.6.1
Post test	- - - -	4.1.2.1.4
Low temperature	3.2.5(b)	4.6.2
Post test	- - - -	4.1.2.1.4
High temperature	3.2.5(c)	4.6.3
Post test	- - - -	4.1.2.1.4
Group II (multiplexers 1 and 2)		
Temperature shock	3.2.5(d)	4.6.4
Post test	- - - -	4.1.2.1.4
Humidity	3.2.5(e)	4.6.5
Performance tests	3.2.1.1 to 3.2.1.6	4.5.1 to 4.5.34
Group III (multiplexers 3 and 4)		
Vibration	3.2.5(f)	4.6.6
Post test	- - - -	4.1.2.1.4
Shock	3.2.5(g)	4.6.7
Performance tests	3.2.1.1 to 3.2.1.6	4.5.1 to 4.5.34

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4.2 Quality conformance inspection. Unless otherwise specified in the contract or purchase order (see 6.2.1), all multiplexer lots offered for acceptance shall be subjected to the quality conformance inspections specified in table II.

TABLE II. Quality conformance inspection.

Examination or test	Requirement paragraph	Method paragraph
Group A (individual inspections)		
Visual and mechanical	3.2.2.1, 3.3.3, 3.3.4	4.4
Burn-in and screening	3.2.5(h)	4.6.8
Performance tests	3.2.1.1 to 3.2.1.6	4.5.1 to 4.5.34
Group B (sampling tests)		
Temperature shock	3.2.5(d)	4.6.4
Vibration	3.2.5(f)	4.6.6
Performance tests	3.2.1.1 to 3.2.1.6	4.5.1 to 4.5.34

4.2.1 Inspection lot. All multiplexer lots offered for acceptance at one time shall be assembled into inspection lots as defined in MIL-STD-105.

4.2.2 Individual inspections. Individual inspections shall be performed prior to sampling tests. Each multiplexer offered for acceptance shall be subjected to the tests of Group A of table II in the sequence specified therein. Any multiplexer that fails to meet the specified requirements shall be rejected and removed from the lot. Additional multiplexers may be added to the lot to provide the contract or purchase order quantity, prior to quality conformance sampling test.

4.2.3 Sampling. Upon completion of individual inspection of 4.2.2, the lot shall be randomly sampled as follows:

Lot size	Sample size
2-50	2
51-150	4
151-300	6
301-500	8

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The sample shall be subjected to the Group B tests of table II. Failure of the multiplexer to pass any of the tests specified shall reject the lot represented.

4.3 Test documentation and test equipment.

4.3.1 Test documentation.

4.3.1.1 Test procedures. When specified in the contract or purchase order (see 6.2.2), test procedures shall be submitted to the procuring activity for approval.

4.3.1.2 Test reports. When specified in the contract or purchase order (see 6.2.2), test reports shall be submitted to the procuring activity for approval. Sample data sheets are shown in figure 3.

4.3.1.3 Certification. The contractor shall have available written certification that the components used as a part of the multiplexer meet the applicable requirements as defined in MIL-STD-109.

4.3.2 Test equipment.

4.3.2.1 Maintenance and calibration. Unless otherwise specified in the contract or purchase order, all test equipment shall be calibrated and maintained in accordance with the provision of MIL-C-45662 (see 6.2.1). When specified in the contract or purchase order, certification of calibration and maintenance shall accompany all reports of preproduction and quality conformance inspections (see 6.2.2).

4.3.2.2 Test equipment accuracy. All test equipment used in the performance of the specified electrical tests shall have an accuracy of at least one-fifth the tolerance for the variable to be measured.

4.4 Visual and mechanical examination.

4.4.1 Visual examination. The multiplexer shall be visually examined to verify that marking and workmanship meet the requirements of 3.3.3 and 3.3.4.

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4.4.2 Physical examination. A complete physical examination shall be performed on the multiplexer to verify that all envelope and interface dimensions and the weight meet the requirements of figure 1.

4.5 Performance tests. Unless otherwise specified herein, the multiplexer shall be operated for test purposes on a +28-Vdc supply and on internal clock (at the highest rate if multiple clock outputs are present). The PROM used for testing shall be set to sample each input once and with a 10-bit resolution, with 10101001010011110011011001010O as the synchronization pattern. If parity is present, parity shall be used as the eleventh bit in each word. The output shall be loaded by 3-kilohm, ±1 percent resistor to a 4.75- to 5.25-Vdc supply and continuously monitored. The filtered output (if present) shall be loaded by a 75-ohm ±1 percent resistor.

4.5.1 Input-voltage range for high-level inputs. Input-voltage range for high-level inputs shall be made as follows:

(a) Set the multiplexer for 10-bit outputs.

(b) Ground all high-level inputs.

(c) Record the output of each channel on figure 3(B).

(d) Apply 5.000 ±0.001 Vdc from a precision source to all high-level inputs.

(e) Verify that at the output all such inputs produce an output of 11111101XX, and record the data on figure 3(B).

To be acceptable, the multiplexer shall meet the requirements of 3.2.1.1.1.1 and 3.2.1.1.1.2.

4.5.2 Input resistance for high-level channels. Input resistance for high-level channels shall be measured as follows:

(a) Connect the multiplexer to produce an output of 10 bits per word.

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(b) Ground the input of a high-level channel, and record the output signal on figure 3(B).

(c) Reconnect the input to ground through a 1-kilohm resistor, and record the output on figure 3(B). The output should not change by a factor greater than 2 bits.

(d) Connect a precision source of 5.00 Vdc to the channel under test and record the output on figure 3(B).

(e) Reconnect the input through a 1-kilohm series resistor and again record the output on figure 3(B). The output should not change by a factor greater than 2 bits.

To be acceptable, all high-level input channels shall meet the requirements of 3.2.1.1.1.3 and 3.2.1.1.1.4.

4.5.3 Back current. Connect a zero-centered ±1 μA meter between any high-level input channel and data ground. Manually clock the multiplexer to the selected channel. Measure the back current and record the data on figure 3(C). To be acceptable, the multiplexer shall meet the requirements of 3.2.1.1.1.4.

4.5.4 Out-of-band voltages. With one high-level analog signal input lead grounded to multiplexer signal ground, apply +29.9 to +30.0 Vdc to three other inputs. The output of the grounded channel shall read 00000011XX; the other shall read 1111111111. Repeat for -29.9 to -30.0 Vdc; output of the grounded channel shall still be 00000011XX, for the others 0000000000. Verify the out-of-band voltages and record data on figure 3(C). To be acceptable, the multiplexer shall meet the requirements of 3.2.1.1.1.5 for all high-level input channels.

4.5.5 Data ground. With the test setup as in 4.5.4, insert a power supply of 40 Vdc between data and power ground. The grounded channel shall still read 00000011XX. Reverse the polarity of the 40-Vdc supply. The grounded channel shall still read 00000011XX, with the two XX values unchanged. Reverse the 40-Vdc supply as in 4.5.4 and repeat the two previous steps. To be acceptable, the multiplexer shall meet the requirements of 3.2.1.1.1.6.

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4.5.6 Low-level inputs, minimum voltage. Apply the minimum voltage specified for low-level channels in 3.2.1.1.2.1 to all low-level channels. Output of each channel shall be 00000001XX to 00000011XX. To be acceptable, the multiplexer shall meet the requirements of 3.2.1.1.2.1.

4.5.7 Low-level inputs, maximum voltage. Apply the maximum voltage specified for low-level channels in 3.2.1.1.2.2 to all low-level channels. Output of each channel shall be 11111100XX to 11111110XX. To be acceptable, the multiplexer shall meet the requirements of 3.2.1.1.2.2.

4.5.8 Common-mode rejection. With the test setup of 4.5.7, apply an alternating-current generator with peak-to-peak voltage equal to the peak-to-peak sensitivity of the selected low-level channel (see 3.2.1.1.2.1 and 3.2.1.1.2.2) between the combination of the positive and the negative input lead of the channel under test and data ground. Set the generator frequency at 10 Hz. Verify and record in figure 3(C) that no more than a single bit change occurs in the channel under test. Repeat for 60 Hz and observe for the same result. Repeat for 100 and 1,000 Hz, and record on figure 3(C) the maximum and minimum outputs. To be acceptable, all low-level channels shall meet the requirements of 3.2.1.1.2.3.

4.5.9 Common-mode voltage range. Replace the alternating-current source used in 4.5.8 with a direct-current adjustable source. No change in the output shall occur beyond the seventh most significant digit (1 part in 128) as the supply is varied from 0 to +10 Vdc and from 0 to -10 Vdc. Verify common-mode voltage range and record data on figure 3(C). To be acceptable, all low-level channels shall meet the requirements of 3.2.1.1.2.4.

4.5.10 Input resistance for low-level channels. Input resistance for low-level channels shall be measured as follows:

(a) Connect the multiplexer to produce an output of 10 bits per word.

(b) Connect a precision voltage source of voltage equal to the zero-scale voltage specified for the low-level channel under test (see 3.2.1.1.2) between the input leads of the channel under test. Record the output of the channel on figure 3(D).

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(c) Connect a 1-kilohm resistor in series with the source and again record the output of the channel on figure 3(D). The output shall not change by a factor greater than 4 bits.

(d) Repeat (b) and (c) for a voltage equal to the full-scale voltage specified for the low-level channel under test and record the data on figure 3(D).

To be acceptable, all low-level input channels shall meet the requirements of 3.2.1.1.2.5.

4.5.11 Back current, low-level channels. With the negative input to a low-level channel connected to data ground, connect a nanoammeter between the positive input and ground, and measure the current. Next, connect the nanoammeter between the positive and negative input terminals (with neither grounded) and measure and record the current on figure 3(D). To be acceptable, all low-level input channels shall meet the requirements of 3.2.1.1.2.6.

4.5.12 Out-of-band voltages, low-level channels. Apply 10 Vdc between the terminals of a low-level input, then remove the power and ground both inputs. The output of this channel shall still meet the requirements specified in 3.2.1.1.2.1 and 3.2.1.1.2.2. Reverse the 10-Vdc power supply and repeat. To be acceptable, all low-level channels shall meet the requirements of 3.2.1.1.2.7.

4.5.13 Recovery time, low-level channels. Apply 10 Vdc as in 4.5.12 between the terminals of a low-level input; connect all others to a precision supply producing a voltage equal to the minimum voltage specified for that channel (see 3.2.1.1.2.1). All channels shall perform normally (same output as in tests of 4.5.6) except the channel to which the 10 Vdc is applied, in both polarities. To be acceptable, all low-level channels shall meet the requirements of 3.2.1.1.2.8.

4.5.14 Input impedance for digital channels. Measure input impedance for digital channels in the following manner:

(a) Connect a variable power supply in series with a variable resistor to the selected digital input.

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(b) Set the resistor to zero.

(c) Increase the voltage of the adjustable supply to the point that the output changes from 0 to 1, and record this voltage on figure 3(E).

(d) Increase the voltage of the adjustable supply to 1.1 times the value obtained in (b).

(e) Increase resistance of the adjustable resistor until the output of the multiplexer for the channel under test falls from 1 to 0 and record the data on figure 3(E).

(f) For the purposes of this test, the input impedance of the high digital channel is 10 times the resistance of the adjustable resistor; record this value on figure 3(E).

(g) Measure the input capacitance with a capacitance meter which uses a test voltage of less than 10 V_{p-p}, and record the value on figure 3(E).

4.5.15 Back current, digital inputs. Place a microammeter between the digital input and signal ground, and record the result on figure 3(E). To be acceptable, all the digital inputs shall meet the requirements of 3.2.1.1.3.2.

4.5.16 Threshold voltage. With an adjustable power supply, determine the voltage at which the input changes state when varying the voltage from -30 to +30 Vdc and back down to -30 Vdc. To be acceptable, all the digital inputs shall meet the requirements of 3.2.1.1.3.3.

4.5.17 Throughput delay, digital inputs. Verify by any suitable means that a signal which changes status up to 2 µs prior to selection of that channel will be transmitted properly. This verification shall take the form of a circuit description if digital storage and transfer are used. To be acceptable, all the digital channels shall meet the requirements of 3.2.1.1.3.4.

4.5.18 Linearity. With the multiplexer set to provide a 10-bit output, apply a ramp voltage of 5 Vdc peak amplitude and 10

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seconds duration to a high-level analog input, and record the output of that channel during the ramp period on figure 3(E). Measure the time interval between 1-bit transitions on the output. The output shall change monotonically upward, with transitions never less than 5 nor more than 20 ms apart. Record data on figure 3(E). To be acceptable, all analog channels shall meet the requirements of 3.2.1.2.4.

4.5.19 Operating voltage range. Perform the test of 4.5.1, 4.5.6, and 4.5.7, with the multiplexer operating on 22 Vdc and 34 Vdc. To be acceptable, the multiplexer shall meet the requirements of 3.2.1.3.1.

4.5.20 Isolation. Measure the resistance between power ground and signal ground leads and record on figure 3(F). To be acceptable, the multiplexer shall meet the requirements of 3.2.1.3.2.

4.5.21 Differential voltage. Perform the tests of 4.5.1, 4.5.6, and 4.5.7 with a 40 Vdc power supply connected between power and data ground. Reverse the polarity of the power supply and repeat the tests. To be acceptable, the multiplexer shall meet the requirements of 3.2.1.3.3.

4.5.22 Current drain. Measure current drawn by the multiplexer during the tests of 4.5.21, and record them on figure 3(F). To be acceptable, the multiplexer shall meet the requirements of 3.2.1.3.4.

4.5.23 Overvoltage. Apply 40 Vdc to the power input leads of the multiplexer for a period of 1 minute, then reduce power to 28 Vdc, and perform the tests of 4.5.1, 4.5.6, and 4.5.7. To be acceptable, the multiplexer shall meet the requirements of 3.2.1.3.5.

4.5.24 Power reversal. Reverse polarity of the power supply used in 4.5.23, and leave connected for a period of 1 minute. Return the power supply to normal polarity and perform the tests of 4.5.1, 4.5.6, and 4.5.7. To be acceptable, the multiplexer shall meet the requirements of 3.2.1.3.6.

4.5.25 Internal clock. Measure the internal clock frequency with a suitable counter and record the data on figure 3(F). To be

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acceptable, the multiplexer shall meet the requirements of 3.2.1.4.1.

4.5.26 Clock load. Connect a 1.5-kilohm resistor from the clock output to ground and verify that a 2.5-V_{p-p} signal is being generated. Reconnect the 1.5-kilohm load to a +5-Vdc source and again verify that a 2.5-V_{p-p} (or greater) signal is being generated. To be acceptable, the multiplexer shall meet the requirements of 3.2.1.4.3.

4.5.27 External clock. Feed the output of a TTL oscillator operating at 10 kilohertz into the external clock input of the multiplexer and verify normal operation. Repeat for frequencies of 10 Hz and 1 megahertz.

4.5.28 External clock loading. Verify by use of a milliammeter that less than 1.67 mA is drawn by the external clock input when the other terminal of the meter is connected to ground. Repeat with the meter connected between the clock input and a +5-Vdc supply. To be acceptable, the multiplexer shall meet the requirements of 3.2.1.4.5.

4.5.29 Clock buffer. When the multiplexer is supplied with an optional clock buffer (see 6.2.1), the contract or purchase order shall specify what testing, if any, shall be performed on those circuits to verify compliance with 3.2.1.4.6.

4.5.30 Output characteristics. Verify compliance of the output format with the output format specified by 3.2.1.5.

4.5.31 Output load. Place a 3-kilohm load between the TTL output and ground, and verify that the output is at least 2.5 V_{p-p}. Repeat for a 3-kilohm load between the TTL output and a +5-Vdc supply. To be acceptable, the multiplexer shall meet the requirements of 3.2.1.5.1.

4.5.32 Output short-circuit protection. With power applied to the multiplexer, short the TTL output to ground for a period of at least 1 minute. Remove the short and verify normal functioning of the output. To be acceptable, the multiplexer shall meet the requirements of 3.2.1.5.2.

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4.5.33 Filtered output. When a filtered output is provided (see 6.2.1), verify that the multiplexer filtered output produces a 2-V_{p-p} (or greater) signal when loaded with 75 ohms to ground. To be acceptable, the multiplexer shall meet the requirements of 3.2.1.6.

4.5.34 Output filter. When a filtered output is provided (see 6.2.1), verify filter frequency by operating the multiplexer with external clock frequencies equal to one-tenth the specified filter frequency, twice the specified frequency, and four times the specified frequency. Output peak-to-peak voltages on a 010 or 101 transition for the twice-specified frequency case shall be 0.6 to 0.8 of the response at the one-tenth rate; output peak-to-peak voltage for the four-times frequency shall be 0.002 to 0.003 times the response at the one-tenth rate. To be acceptable, the multiplexer shall meet the requirements of 3.2.1.6.

4.6 Environmental.

4.6.1 Temperature-altitude. The multiplexer shall be tested in accordance with MIL-STD-810C, Method 504.1, Procedure I, Steps 2 and 3 only. The temperature for both Steps 2 and 3 shall be -40°C (-40°F), and the altitude for Step 3 shall be 110,000 feet. Operational and performance checks are not required during exposure; however, external power shall be applied in accordance with the procedures of Steps 2 and 3. Following Step 3, the tests of 4.1.2.1.4 shall be performed. To be acceptable, the multiplexer shall meet the requirements of 3.2.5(a).

4.6.2 Low temperature. The multiplexer shall be tested in accordance with MIL-STD-810C, Method 502.1, Procedure I. Storage temperature shall be -54°C (-65.2°F) and storage duration shall be 16 hours. Operating temperature shall be -40°C (-40°F), and Step 4 duration shall be 16 hours. Step 5 shall consist of the application of external power for a period of 2 hours. Step 7 shall consist of the tests of 4.1.2.1.4. To be acceptable, the multiplexer shall meet the requirements of 3.2.5(b) and 3.2.5(c).

4.6.3 High temperature. The multiplexer shall be tested in accordance with MIL-STD-810C, Method 501.1, Procedure I, except that Step 3 shall be 24 hours duration. Storage temperature shall be +85°C (+185°F) and storage duration shall be 4 hours. Operating temperature shall be +80°C (+176°F). Step 5 shall consist of the

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application of external power for a period of 2 hours, while maintaining the temperature at +85°C (+185°F). Step 7 shall consist of the tests of 4.1.2.1.4. To be acceptable the multiplexer shall meet the requirements of 3.2.5(b), and 3.2.5(c).

4.6.4 Temperature shock. The multiplexer shall be tested in accordance with MIL-STD-810C, Method 503.1, Procedure I, except that Step 2 temperature shall be -40°C (-40°F). Step 9 shall consist of the tests of 4.1.2.1.4. To be acceptable, the multiplexer shall meet the requirements of 3.2.5(d).

4.6.5 Humidity. The multiplexer shall be tested in accordance with MIL-STD-810C, Method 507.1, Procedure I. Step 7 shall consist of the tests of 4.1.2.1.4. To be acceptable, the multiplexer shall meet the requirements of 3.2.5(e).

4.6.6 Vibration. The multiplexer shall be tested in accordance with MIL-STD-810B, Method 514.1, Procedure II, Curve D for part 1, Curve S for part 2, and Curve AK for part 3. Unless otherwise specified in the contract or purchase order (see 6.2.1), combined temperature and vibration tests shall be performed on all preproduction units. Combined temperature and vibration functional testing shall be performed at each temperature extreme of 3.2.5(c) following a 2-hour soak. Vibration isolators shall not be used. To be acceptable, the multiplexer shall meet the requirements of 3.2.5(f).

4.6.7 Shock. The multiplexer shall be tested in accordance with MIL-STD-810C, Method 516.2, Procedure IV, Flight Vehicle Equipment, Figure 516.2-1 100 g amplitude and a time duration of 6 ms. When an operation is required, the tests of 4.1.2.1.4 shall be performed. Shock isolators shall not be used. To be acceptable, the multiplexer shall meet the requirements of 3.2.5(g).

4.6.8 Burn-in and screening. Each multiplexer shall be subjected to a minimum of 6 cycles of burn-in and screening as shown in figure 2. To be acceptable, the multiplexer shall meet the requirements of 3.2.5(h).

4.6.8.1 Temperature cycling. Temperature cycling shall be as shown in figure 2. A thermal survey shall be performed prior to testing to determine the time required for the chamber air temperature to stabilize at the high and low temperatures indicated. The duration of 1 cycle shall be the sum of the stabilization times and the 2-hour dwell time at each temperature extreme.

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4.6.8.2 Vibration subcycle. The vibration power spectral density shall be as shown in figure 4. Vibration shall be of 15 minutes duration for each exposure. Each exposure shall be in one axis only, with successive exposures in alternate axes, so that the item will have been vibrated in each of three mutually perpendicular axes at least once during the screening test. The item may be removed from the temperature chamber for the performance of the vibration test, and the vibration exposure shall commence within 15 minutes of removal from the chamber.

4.6.8.3 Operation of the multiplexer. As indicated in figure 2, the multiplexer shall be operated during the heating subcycle. Operation shall consist of the application of nominal external power. Performance shall be monitored in a manner to detect any cessation of function, but parameter limits need not be verified during this operation.

4.6.9 Failure analysis. Should the multiplexer fail during testing to meet any of the requirements specified herein, the test shall be interrupted until the multiplexer has been examined. If a decision is made to discontinue testing of the multiplexer, a complete failure analysis shall be performed and, when specified in the contract or purchase order, a report shall be prepared and forwarded to the procuring activity (see 6.2.2).

4.7 Packaging, packing, and marking. Examination shall be made to ascertain that packaging, packing, and marking are in accordance with Section 5.

5. PREPARATION FOR DELIVERY.

5.1 Preservation and packaging.

5.1.1 Level C. The multiplexer shall be preserved and packaged in a manner to afford protection against damage during direct shipment from the supply source to the first receiving activity for immediate use.

5.2 Packing.

5.2.1 Level C. The multiplexer shall be packed in a manner to ensure carrier acceptance and safe delivery at destination.

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5.3 Marking. In addition to any special markings required by the contract or purchase order (see 6.2.1), unit packages, intermediate packages, and shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES.

6.1 Intended use. The multiplexer will be used as part of various airborne telemetry systems to obtain in-flight performance data.

6.2 Ordering data. Procurement documents should specify the following:

6.2.1 Procurement requirements.

(a) Title, number, and date of this specification.

(b) Analog and digital input characteristics (see 3.2.1.1).

(c) Input voltage range for high-level analog inputs if different from 3.2.1.1.1.

(d) The output from zero V input if different from 3.2.1.1.1.1.

(e) The output from 5.000 V input if different from 3.2.1.1.1.2.

(f) Input voltage range for low-level analog inputs (see 3.2.1.1.2).

(g) Voltage range for discrete inputs if different from 3.2.1.1.3.

(h) Resolution of the analog-to-digital conversion if different from 3.2.1.2.2.

(i) Conversion speed if different from 3.2.1.2.3.

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(j) The amperage the multiplexer can draw if different from 3.2.1.3.4.

(k) Requirement for frequency of internal clock (see 3.2.1.4.1).

(l) Requirement for clock buffer circuit (see 3.2.1.4.6).

(m) Requirement for a single parity bit at the end of each data word (see 3.2.1.5).

(n) Requirement for a filtered output (see 3.2.1.6, 4.5.33, and 4.5.34).

(o) Conformance to documents if different from 3.2.2.1.

(p) Temperature and vibration tests if different from 4.6.6.

(q) Maximum permissible electromagnetic radiation emission levels (see 3.3.2).

(r) Electromagnetic radiation testing if different from 3.3.2.

(s) When a preproduction sample is required, testing and approval will be under the appropriate provisions of 7-104.55 of the Armed Services Procurement Regulations. The preproduction sample should consist of 4 units. The contracting officer should include specific instructions in all procurement instruments, regarding arrangements for examinations, tests, and approval of the preproduction sample.

(t) Responsibility for performance of inspection requirements if different from 4.1.1.

(u) Testing facilities if different from 4.1.1.

(v) Quality conformance inspections if different from 4.2.

(w) Calibration and maintenance of test equipment if different from 4.3.2.1.

(x) Clock buffer circuit testing, if required (see 4.5.29 and 6.2.1(l).

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(y) Special markings (see 5.3).

(z) Output signaling rate (see 3.1).

6.2.2 Data requirements. When this specification is used in a procurement which incorporates a DD Form 1423 and invokes the provisions of 7-104.9(n) of the Armed Services Procurement Regulations, the data requirements identified below will be developed as specified by an approved Data Item Description (DD Form 1664) and delivered in accordance with the approved Contract Data Requirements List (DD Form 1423) incorporated into the contract. When the provisions of ASPR-7-104.9(n) are not invoked, the data specified below will be delivered by the contractor in accordance with the contract requirements. Deliverable data required by this specification is cited in the following paragraphs:

Paragraph	Data Requirement	Applicable DID
3.2.1.7	PROM Programming Instructions	DI-M-4712
3.3.2	Test Report (Electro-magnetic Radiation Tests)	DI-T-2072
4.3.1.1	Test Procedures	DI-T-5204
4.3.1.2	Test Reports	DI-T-2072
4.3.2.1	Certification	UDI-A-2364B
4.6.9	Failure Analysis Report	UDI-R-21141A

(Copies of data item descriptions required by the contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

Preparing activity:
Navy(AS)
Project number 5821-N142

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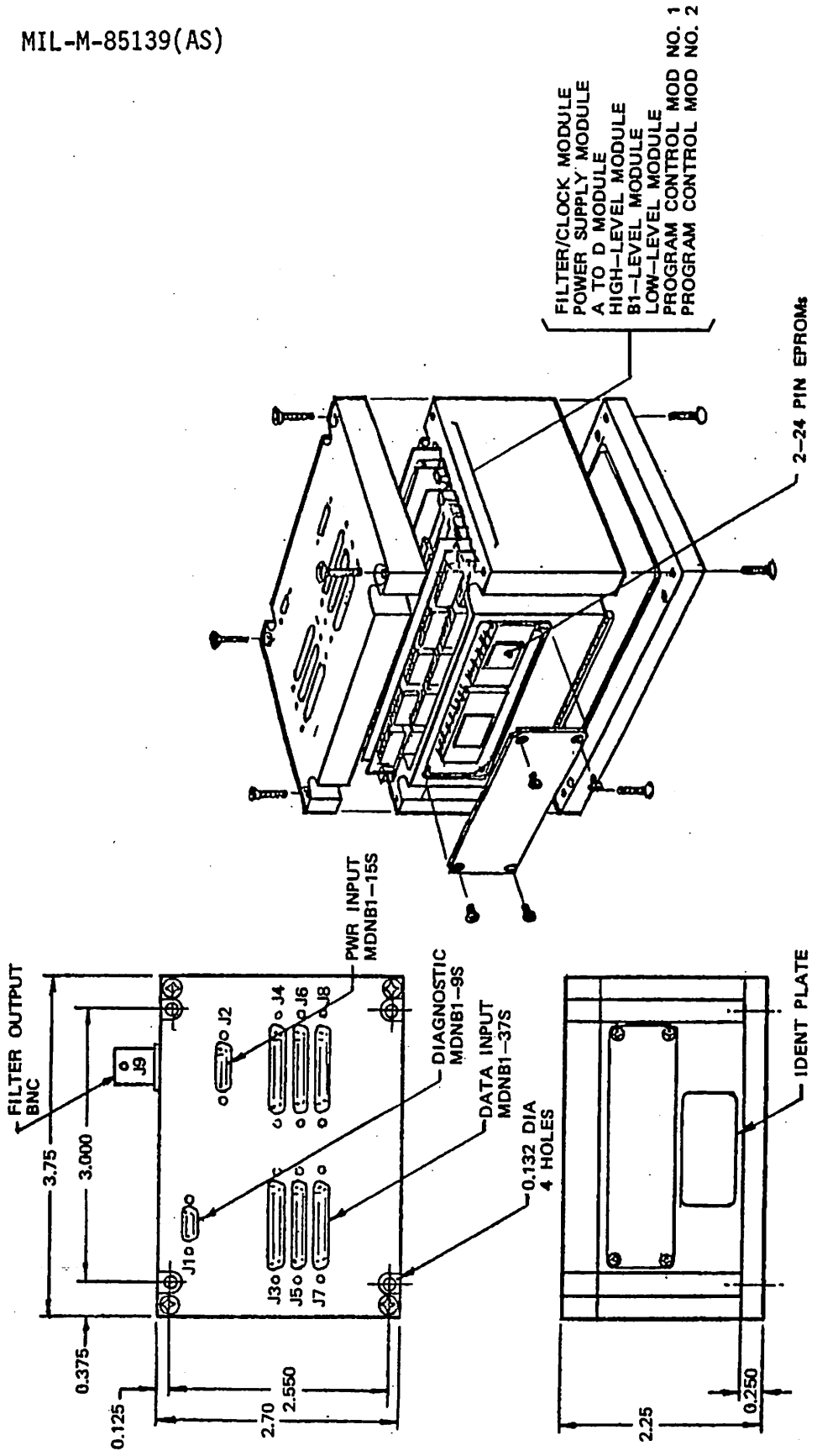
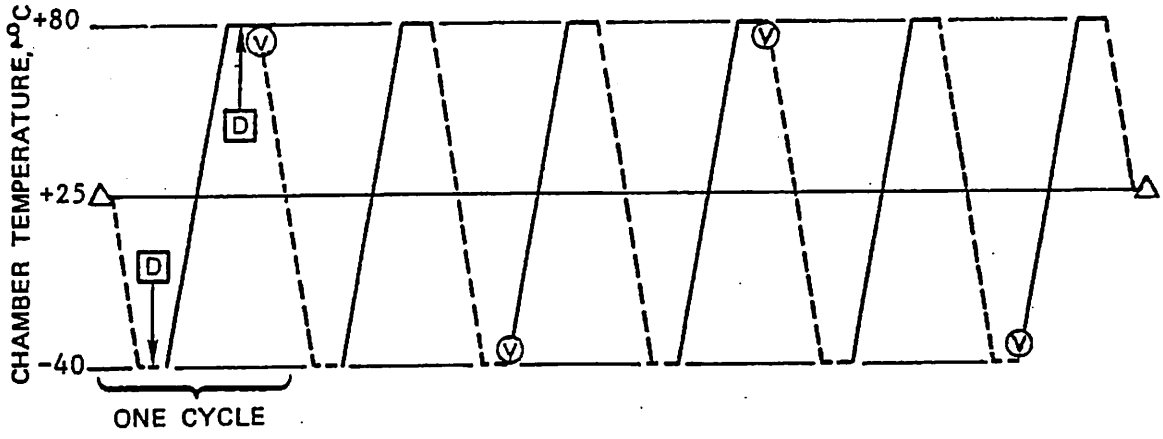


FIGURE 1. Multiplexer envelope.

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LEGEND

△ = COMPLETE FUNCTIONAL PERFORMANCE TEST
[D] = DWELL TIME AT TEMPERATURE EXTREMES
Ⓥ = VIBRATION EXPOSURE SUBCYCLE
---= EQUIPMENT NONOPERATING (COOL DOWN)
——= EQUIPMENT OPERATING (HEATING SUBCYCLE)

NOTES:

1. A COMPLETE PERFORMANCE TEST SHALL BE CONDUCTED PRIOR TO THE SCREENING TEST AND AT THE COMPLETION OF THE SIXTH CYCLE.
2. THE LAST TWO CYCLES SHALL BE FAILURE FREE – IF FAILURES OCCUR, REPAIRS SHALL BE PERFORMED, AND THE UNIT SUBJECTED TO A MINIMUM OF TWO ADDITIONAL CYCLES.
3. EQUIPMENT PERFORMANCE SHALL BE CONTINUOUSLY MONITORED DURING THE HEATING SUBCYCLES SUFFICIENTLY TO DETECT CESSATION OF FUNCTION.
4. DWELL TIME, AFTER CHAMBER STABILIZATION SHALL BE 2 HOURS MINIMUM.
5. TEMPERATURE CHAMBER SHALL BE ADEQUATE TO PROVIDE A RATE OF TEMPERATURE CHANGE OF 2°C/MINUTE MINIMUM.

FIGURE 2. Burn-in and screening profile.

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QUALITY ASSURANCE DATA SHEETS

FOR

MULTIPLEXER

MFD. BY _____

CONTRACT NO. _____

LOT NO. _____

SERIAL NO. _____

(A)

FIGURE 3. Sample QA data sheet.

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Input voltage range, high-level inputs (see 4.5.1)	
Ground high-level inputs	
Output (00000010XX)* Verify ( ) _____ (all high-level channels)	
Test voltage 5.000 ±0.001 Vdc	
Output (11111101XX)* Verify ( ) _____ (all high-level channels)	
*Where X is either a 0 or a 1	
Input resistance, high-level channels (see 4.5.2)	
Output of test channel, grounded	_____
Output of test channel, grounded through 1-kilohm resistor	_____
Output of test channel, 5,000 Vdc input	_____
Output of test channel, 5,000 Vdc input through 1-kilohm resistor	_____

Verified _____

Date _____

(B)

FIGURE 3. Sample QA data sheet. - Continued

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Back current (see 4.5.3)	
Back current, high-level channels _____ nA at 25°C	
Out-of-band voltages (see 4.5.4)	
Out-of-band voltages	Verify ( ) _____
Common-mode rejection (see 4.5.8)	
Common-mode rejection at 10 Hz	Verify ( ) _____
Peak-to-peak variation at 100 Hz _____ bits	
Peak-to-peak variation at 1 kilohertz _____ bits	
Common-mode voltage range (see 4.5.9)	
Common-mode voltage range	Verify ( ) _____

Verified _____

Date _____

(C)

FIGURE 3. Sample QA data sheet. - Continued

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Input resistance, low-level channels (see 4.5.10)	
Output of test channel, specified zero-scale voltage	_____
Output of test channel, specified zero-scale voltage through 1-kilohm resistor	_____
Output of test channel, specified full-scale voltage	_____
Output of test channel, specified full-scale voltage through 1-kilohm resistor	_____
Note: Use lowest voltage for grounded conditions. Use highest voltage for 5,000 Vdc input conditions.	
Back current, low-level channels (see 4.5.11)	
Back current _____ nA at 25°C	

Verified _____

Date _____

(D)

FIGURE 3. Sample QA data sheet. - Continued

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:49Z
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Input impedance, digital channels (see 4.5.14)
Threshold voltage _____ at 25°C Calculated resistance _____ ohms Input capacitance _____ pF
Back current, digital inputs (see 4.5.15)
Back current, digital channels _____ nA at 25°C
Linearity (see 4.5.18)
Minimum time between steps _____ ms Maximum time between steps _____ ms

Verified _____

Date _____

(E)

FIGURE 3. Sample QA data sheet. - Continued

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Isolation (see 4.5.20)
Ground isolation _____ ohms
Current drain (see 4.5.22)
Current at 22 Vdc _____ mA Current at 34 Vdc _____ mA
Internal clock (see 4.5.25)
Clock frequency _____

Verified _____

Date _____

(F)

FIGURE 3. Sample QA data sheet. - Continued

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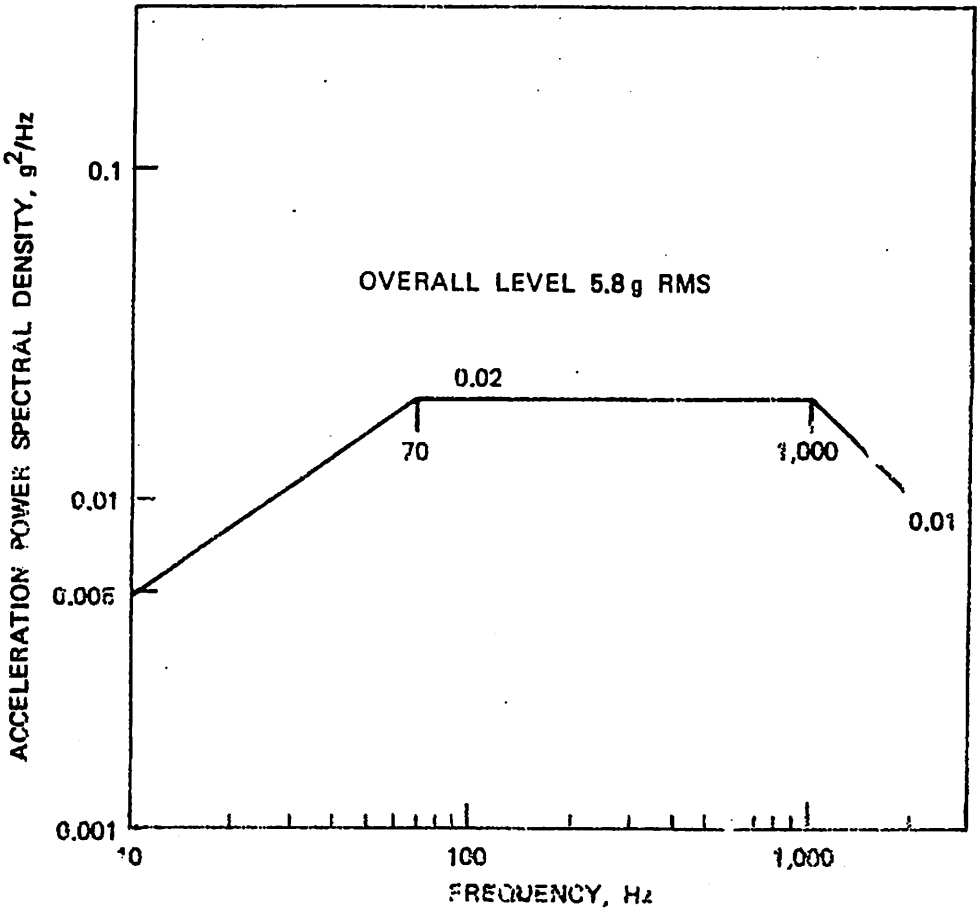


FIGURE 4. Vibration profile, screening.

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